

4

Server AI

***Owens:** FastAPI + Pydantic, VLM/LLM integration, prompt/action schema, context validator.*

Role 4 of 6 · SERVER

HW Hardware access

Heavy only if self-hosting an open-weight model for the offline-capable stretch goal. Cloud API use during SIH needs no local GPU.

Share Laptop B with Integration, booked in advance, only for self-hosted-model tests. Default to a cloud API to stay out of the hardware queue entirely.

P0 Phase 0 — now → 20 Sep 2026 (lock the idea)

- Pick the cloud VLM/LLM API to use during SIH; smoke-test it on a cloud notebook against a hand-built sanitized-context example.
- Freeze a first draft of the **structured action schema** (click/type/scroll/navigate/keypress/wait + type_secret) together with Extension — this blocks their validator work, so do it early.

P1 Phase 1 — build the MVP (20 Sep → shortlist)

- FastAPI + Pydantic server with a sanitized-context endpoint.
- Context Validator: schema-check incoming payloads, reject unexpected raw fields — this doubles as your side of the privacy proof.
- Prompt design so the model reasons over placeholders ([EMAIL_01], [PERSON_01]) without ever seeing or requesting real values.
- Action Candidate Generator + Action Validator: constrain model output to the frozen action schema before anything is returned to the client.

P2 Phase 2 — rehearse & harden (shortlist → Dec finale)

- Load/latency-test the server path alongside Integration's benchmark sweeps.
- Confirm the “server received zero raw canary values” proof from the server side of the request log.

DEP Dependencies

YOU DEPEND ON	Privacy Guard (sanitized context format), Extension (what actions the executor actually supports).
DEPENDS ON YOU	Extension (needs your action-response format to build the validator), PVM (needs a first correct server-reasoned action before anything can be memorized).

Full team plan, hardware-booking rules, and the master phase timeline: see the companion *PS26171 Team Workflow & Hardware Plan* document.